

From Raw Data to Reports: AI-Assisted Workflow for Routine Pharmacokinetic/Pharmacodynamic (PKPD) Analyses

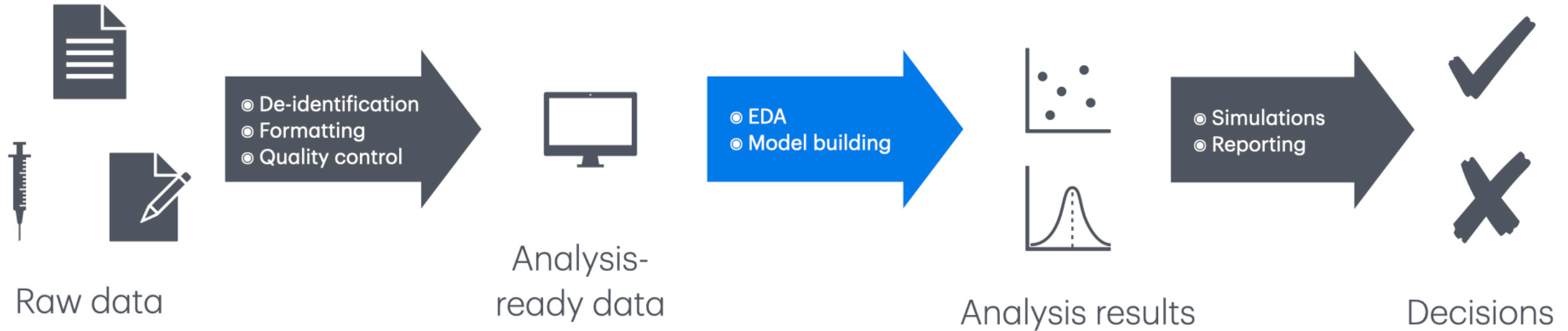
The “routine” workflow



Preparing analysis-ready data



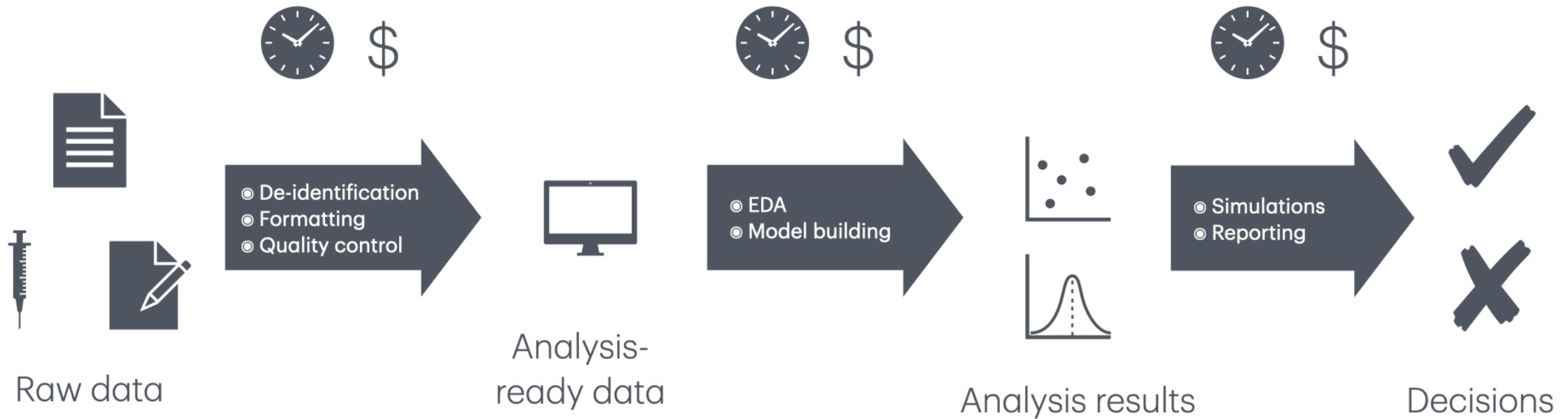
Iterative model building



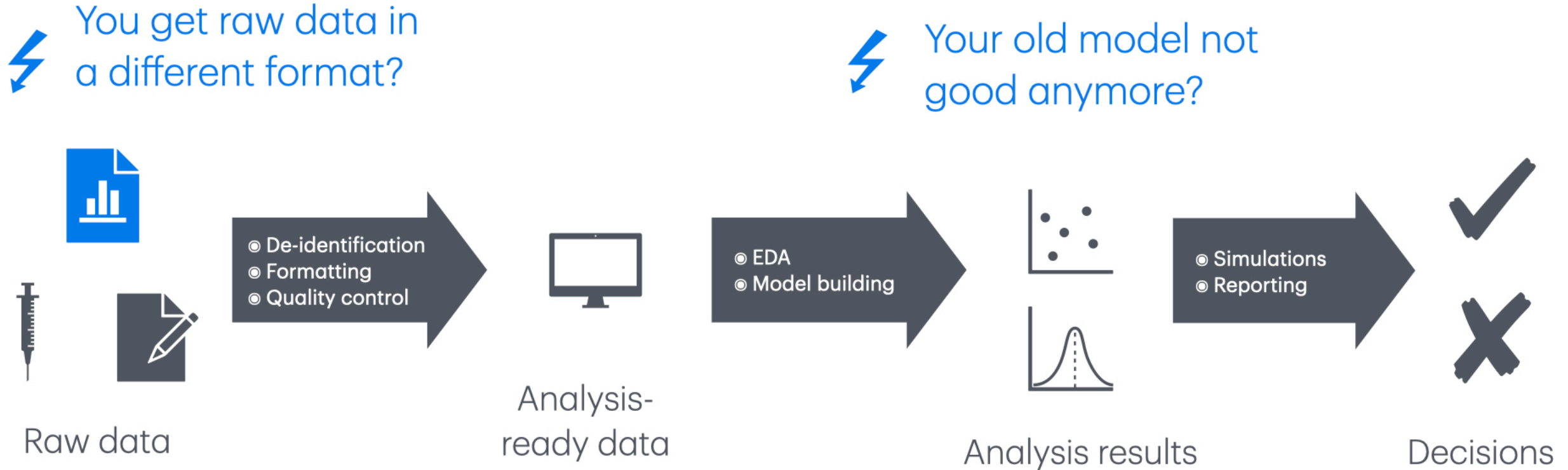
Downstream tasks and reporting



The workflow can take months, costing time and money

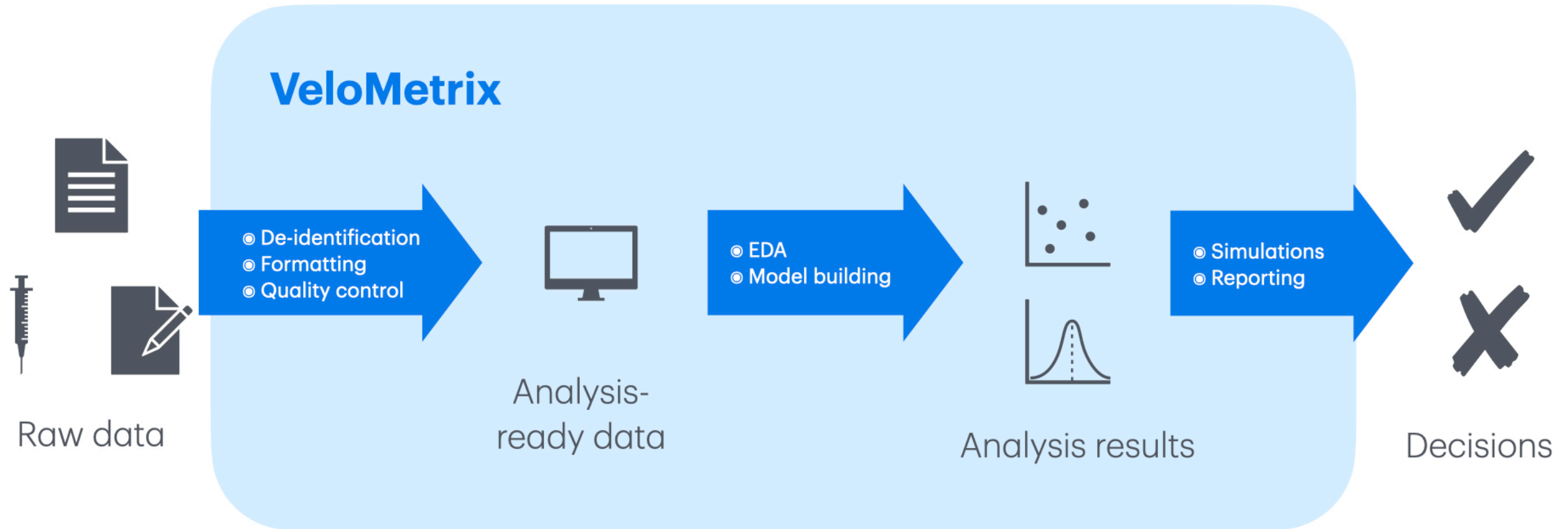


A new data batch restarts the workflow



➡ Need to repeat the workflow ⌚ \$

VeloMetrix aims to automate the workflow



- STUDY
- Studies

Study Dashboard

Study Documents
- DATA
- Data Import

Data Preparation

Data Review

NCA
- MODELING
- Model Building

Results summary

Subject VPCs

Parameter estimates

QOIs

Covariate Effects

Simulator
- OTHER
- Model Library

Settings

STUDY

Study Dashboard | Warfarin PK/PD

See what is happening in the selected study and what happened most recently.

Active std_01KZBWSC1RVJH0GT610CD8DXV5

Upload data →

Archive study

DATA VERSIONS

3

Available

View versions →

UPLOADS

2

0 need input · 0 in progress

View uploads →

MODEL BUILD RUNS

5

0 in progress · 0 with issues

View latest →

Study information

Study name	Warfarin PK/PD
Treatment	Warfarin
Study type	SAD
Phase	Phase I
Description	-

Event history

Showing page 1 with 50 events. More events are available.

All events

Audit (user only)

nca_completion

succeeded

user · Juho Timonen

nca_analysis: nca_01M0GCRDWZP9QY6WPSGJXK3N3B

NCA completed successfully.

Aug 20, 11:10 PM GMT+3

nca_creation

succeeded

user · Juho Timonen

nca_analysis: nca_01M0GCRDWZP9QY6WPSGJXK3N3B

Aug 20, 11:10 PM GMT+3

Data preparation

De-identification · formatting · quality control



Data Import | Multimarker study

Drop your data files. After upload, data preparation starts and asks any clarifying questions it needs.

Upload data

 **Security:** files are encrypted in transit (HTTPS/TLS) and at rest, stored in your workspace's private bucket with access controlled by your workspace authentication.

STUDY	TREATMENT
Multimarker study	Multimarker
std_01M0GE6H1DBHEKCTY0BKMJB414	tmt_01M0GE6GW3WDRQ9PX1WF3G516F

Version description

first version

Brief description

Brief description of the uploaded data

Files *

Drag files here, or click to browse

CSV, TSV, TXT, Excel, SAS7BDAT, XPT, JSON, Parquet, or a single ZIP

Max 100 MB per file, 500 MB per upload. ZIP uploads currently fail if any file inside has an unsupported type.

Configuration (optional, advanced)

Attach a hand-authored gecodata project.yaml to format your data deterministically (warm start), skipping the automatic config step. Leave this empty to let the system infer a configuration from your files (default).

Add config file

Upload

How VeloMetrix handles identifiable data



Without VeloMetrix: repeat de-identification and validation for every delivery.



Study-data batches arrive in many shapes

CSV · Excel · SAS/XPT · Parquet

**One file or
many**

**Different
names**

Different units

**Planned and
actual times**

The next delivery can change the layout again.



Messy exports become a **VeloMetrix** **DataVersion**

Study exports

Different files,
columns, codes,
and units

→ Formatting configuration

Supply a conf file **OR**
work with LLM
assistant

→ **VeloMetrix** formatter

Applies the
configuration
and runs rule-
based checks

→ **VeloMetrix** **DataVersion**

Standardized
tables plus analysis
metadata



Formatting conversation

Data-prep agent

Starting data preparation — inspecting the archive and identifying domain sources.

Data-prep agent

Analyzing data structure — found single file with dosing, PK, and PD observations; checking for PD markers beyond the PK analyte.

Data-prep agent

Data lacks critical metadata (units, route, PD marker definitions). Need to ask for these values before proceeding with config.

Data-prep agent

Drafted ask_user questions for units, route, and PD marker metadata — handing to the user for input.

pk_concentration_unit
mg/mL

ee2a32b241ea10d7da549605bbf35bef732dcda554cc3f5787f655e7208a86b0

dose_amount_unit
mg

ee2a32b241ea10d7da549605bbf35bef732dcda554cc3f5787f655e7208a86b0

route_of_administration
oral

ee2a32b241ea10d7da549605bbf35bef732dcda554cc3f5787f655e7208a86b0

Inside a VeloMetrix DataVersion

Standardized study tables

Subjects

ID	Weight	Group
101	72 kg	B

PK

ID	Time	Conc.
101	2 h	8.4 ng/mL

Dosing

ID	Time	Dose / route
101	0 h	100 mg oral

PD

ID	Time	Marker / value
101	2 h	A: 38%

Analysis metadata

Field roles and types

Canonical units

Eligibility and exclusions

Marker type →
observation model

Available time bases

Dosing route



Inspect the prepared data version before kicking off an analysis.

Data version f45341 (uploaded Aug 20, formatted Aug 20) · latest

Overview

Tabular

Graphical

Explanation

Archive data version

TOTAL SUBJECTS

12

SUBJECTS WITH USABLE PK DATA

11

on observed time basis

ADMINISTRATION ROUTE

Oral

AVAILABLE FOR PKPD MODELING AND NCA

Yes

Data version metadata

PD markers (2)

NAME	TYPE	UNIT	MODELING
pdcont	continuous	unitless	Eligible
pdresp	continuous	unitless	Eligible

Covariates (3)

NAME	TYPE	UNIT / CATEGORIES	IMPORTANT
weight	positive_continuous	-	Yes
age	positive_continuous	-	Yes
sex	categorical	female · male	Yes

Dose groups (1)

DISPLAY NAME	NAME	REGIMENS	SUBJECTS
1.5 mg/kg	1.5 mg/kg	1.5 mg/kg	12

Time bases

Time bases in a VeloMetrix DataVersion

Observed

Recorded dose and sample times

Nominal

Protocol-scheduled dose and sample times

Nominal from observed dose

Recorded dose time + scheduled post-dose offset

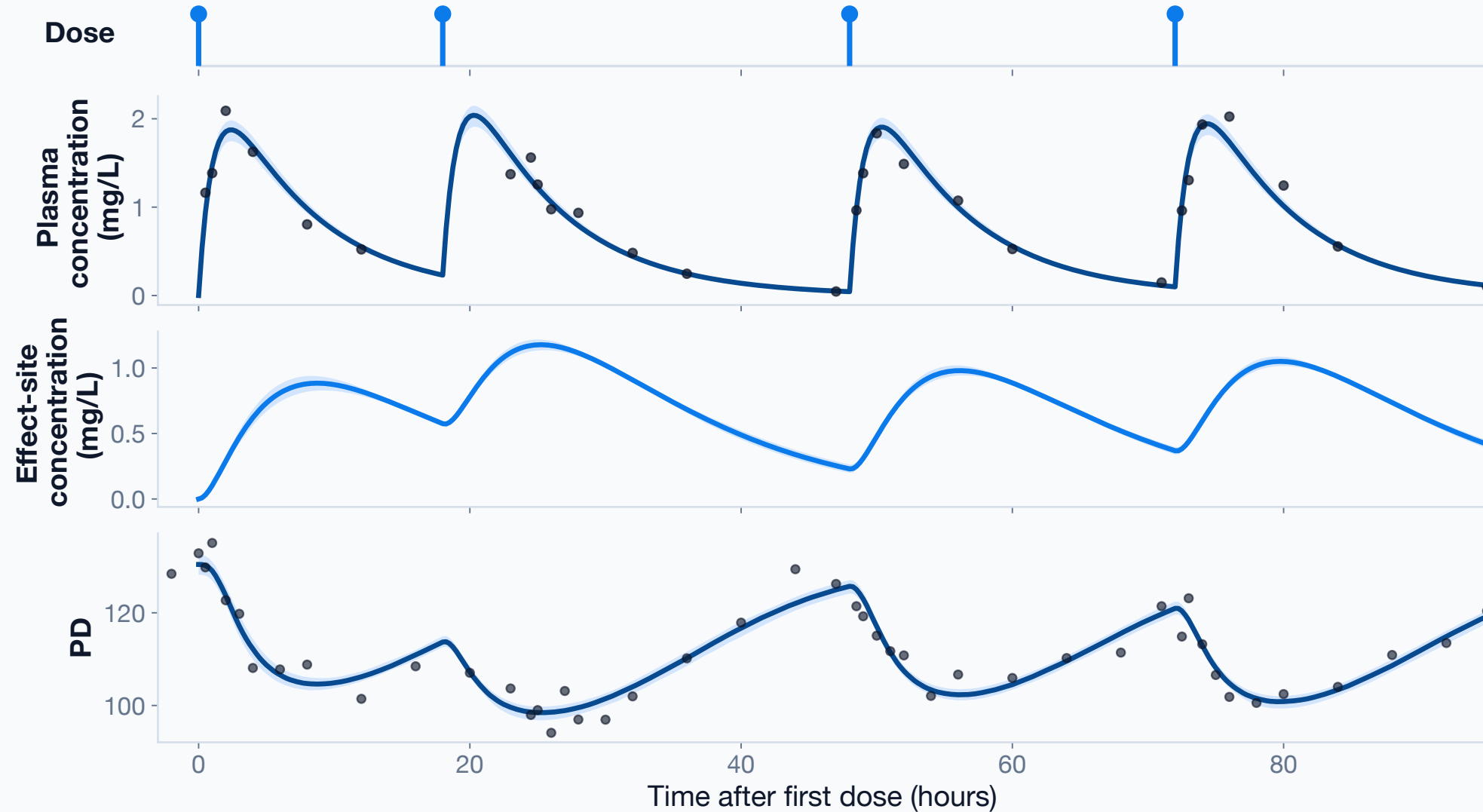


Model building

Typical problems in Bayesian PKPD modeling



PKPD model with an effect compartment



The “routine” manual workflow (Bayesian PKPD)

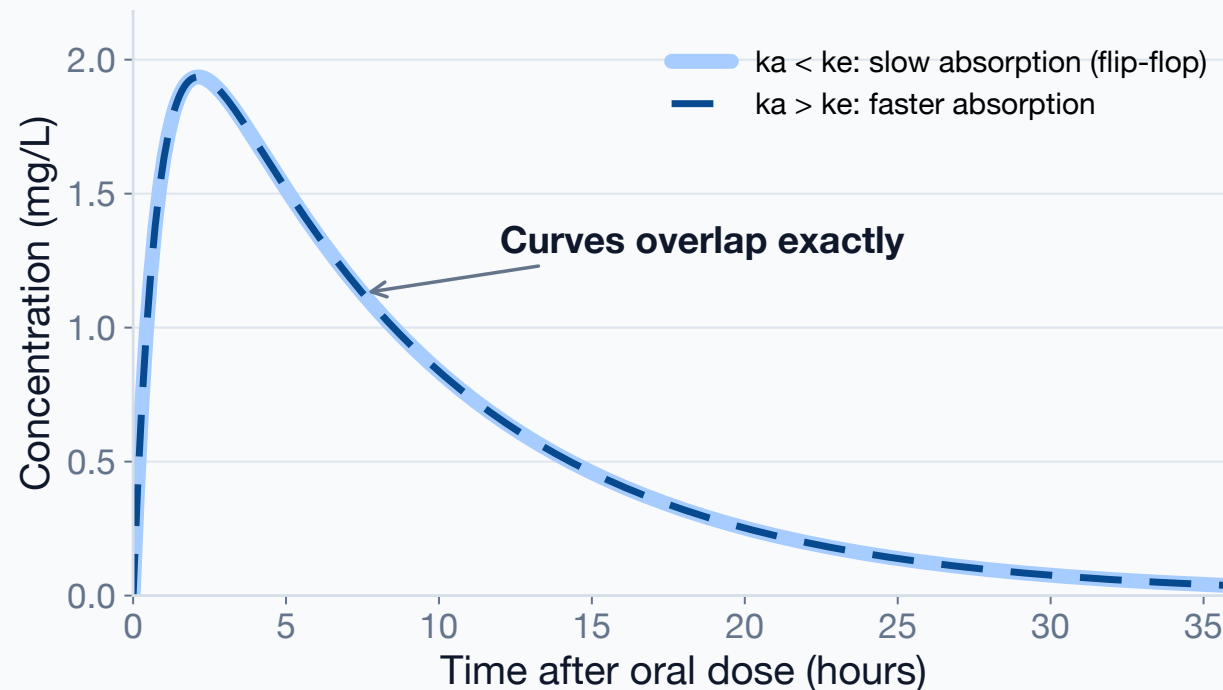
- Try some compartmental PK model
- Make all parameters (like absorption/elimination rates) subject-specific
- Include effects of some covariates that might be important
- Jointly fit a standard PD model like Emax
- Realize you have trouble making the inference converge or the fit is bad
- Simplify or iterate on model structure



Flip-flop in one-compartment oral PK

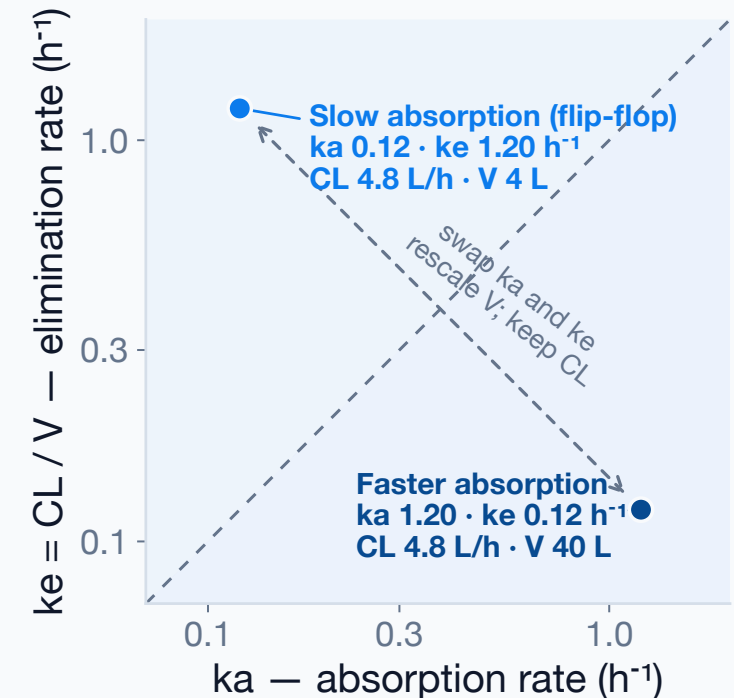


One oral concentration profile



Same $C(t)$, same oral-data likelihood

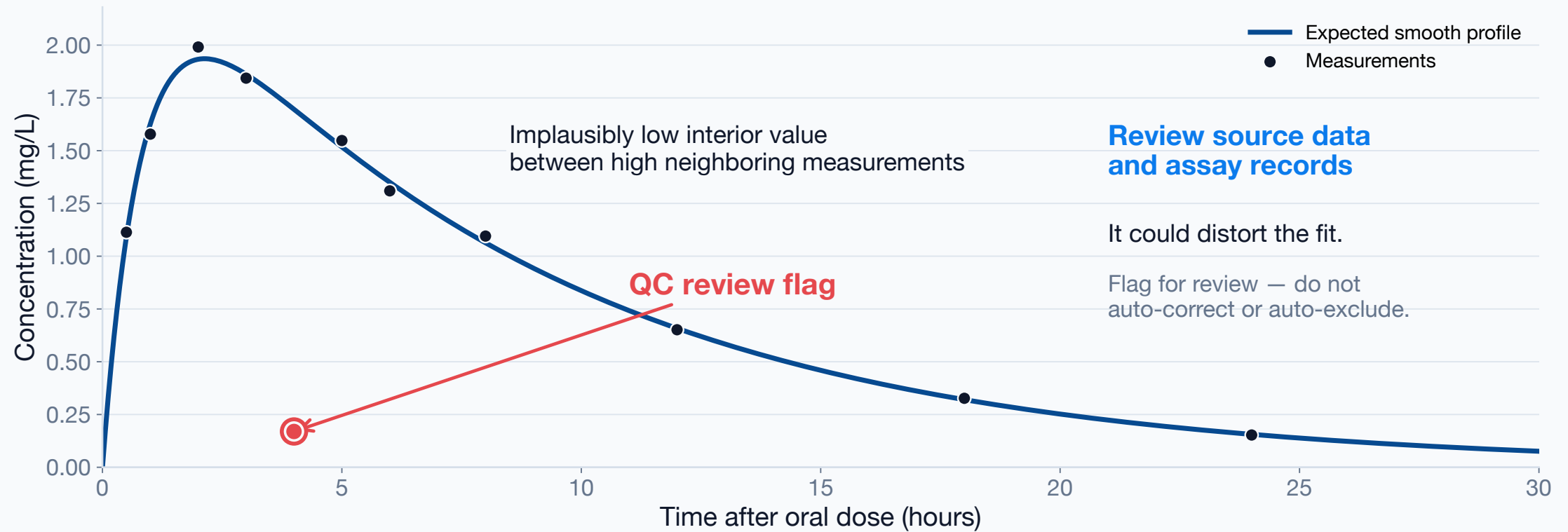
One likelihood, two parameter modes



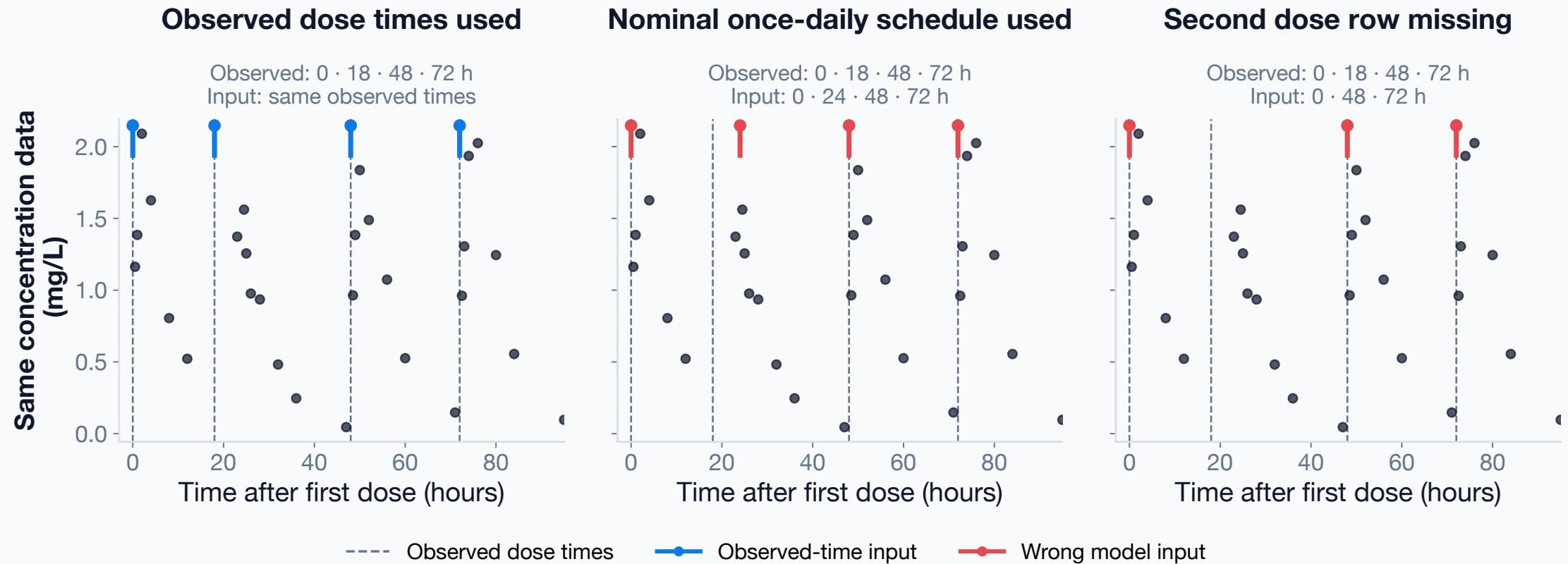
Separate modes can impair mixing and convergence.

Priors can favor one; constraining $k_a > k_e$ excludes flip-flop.

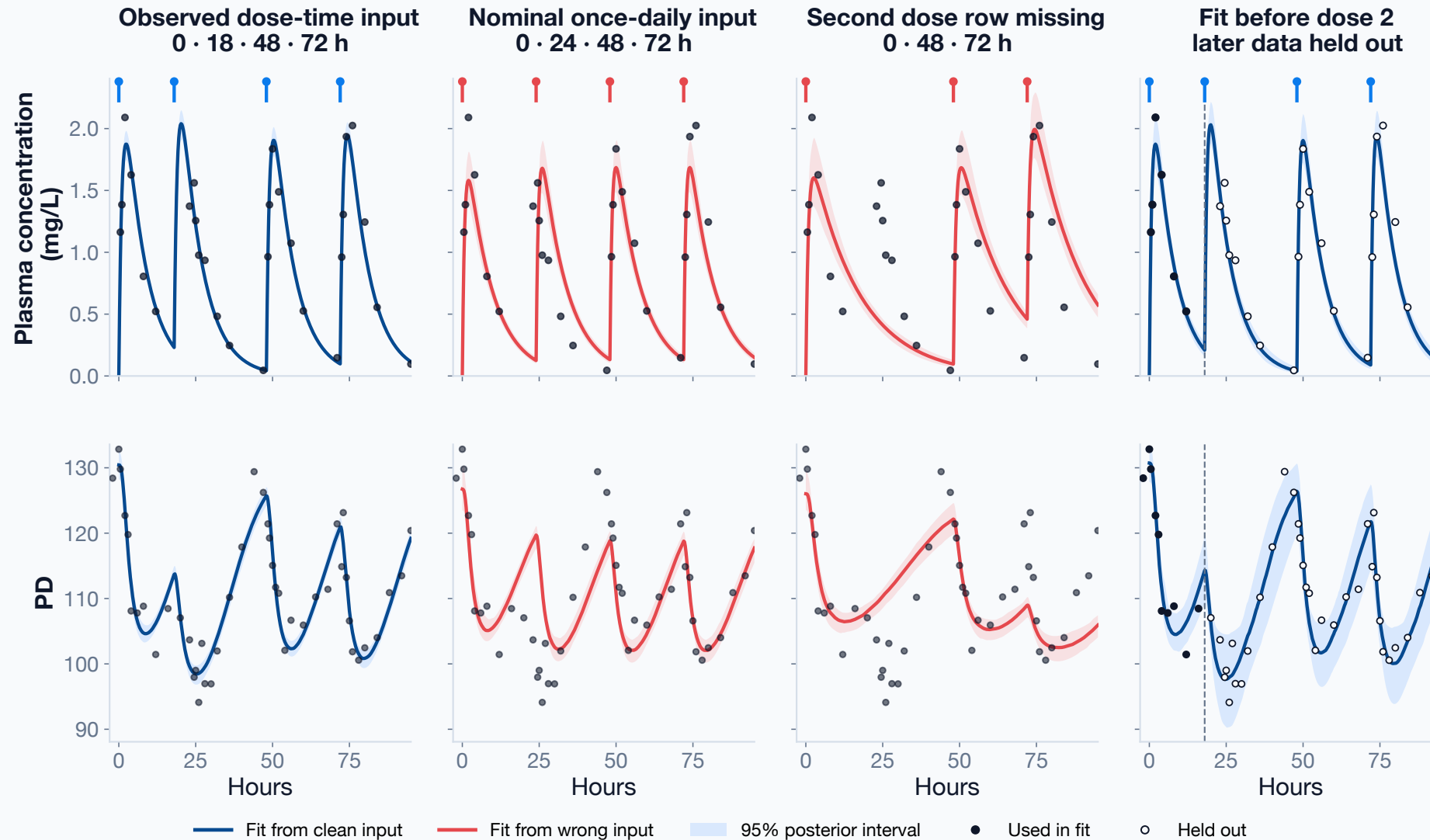
Importance of quality control



Importance of dosing time accuracy



Wrong dose inputs bias PK and PD inference



Model building

VeloMetrix Model Builder



How to build a “foolproof” automatic PKPD Model Builder?

- Needs to work with minimal user input
- Needs to be hard for the user to mess up
- User does not care about Bayes or model details
- Only the end result matters: something that helps them make a decision related to drug development



VeloMetrix Bayesian PKPD Model Builder

- Mimics the iterative manual model building workflow
- Works on the subset of data that passes QC and eligibility checks
- Focus on standard models that cover most routine use cases; prioritize speed and reliability over flexibility
- Search over compartmental PK model structures, then add covariates, then add PD
- Focus on trying to get MCMC to converge for some version each tried model (simplifying fallback versions)



- STUDY
- 

Studies
- 

Study Dashboard
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Study Documents

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NCA

- MODELING
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Model Building
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Results summary
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QOIs
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Covariate Effects
- 

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Model Library
- 

Settings

MODELING

Model Building

Configure and start automatic model build runs for the selected data version.

Data version

f45341 (uploaded Aug 20, formatted Aug 20) · latest

▼

New model build run

Selection

1 time basis available

Choose the PD markers, time basis, and allowed covariates for this run. Subjects missing required inputs are excluded, and covariates that cannot be modeled are unavailable.

PD markers

Select modeled markers and set whether drug exposure is expected to increase or decrease the marker level.

☒ pdcont (pdcont)

Level expected to decrease with drug

▼

☒ pdresp (pdresp)

Level expected to increase with drug

▼

Covariates

Select the covariates to allow in this run. No covariates are selected by default.

☒ weight

IMPORTANT

☒ age

IMPORTANT

☒ sex

IMPORTANT

Time basis

☒ observed

RECOMMENDED

11 PK-eligible subjects on this basis.

Cancel

Preview and start run

Model-building assumptions

Structures

Route-compatible one- and two-compartment PK; effect-compartment Emax PD

Observation models

Selected from marker type in the DataVersion

Hierarchy

Every active PK parameter starts with between-subject variability

Covariates

Every **usable** included covariate enters every active PK parameter

Priors

Model-specific; PD scales use observed response and fitted exposure

Identification

Oral first-order models enforce $k_a > k_e$



Modeling population

This result is the authoritative record of the population used by the model build run.

8 subjects were included. 4 subjects were excluded.

WHY SUBJECTS WERE EXCLUDED

Reasons are separated into data-version PK eligibility and requirements introduced by the selected modeling options. Each subject is counted only once in the excluded total.

Entire subject not PK-eligible

On the selected time basis, the DataVersion marks each listed subject as not PK-eligible. This excludes the whole subject before modeling options are applied.

Entire subject is not PK-eligible

SUBJECTS 12

Excluded by selected modeling options

These reasons are subject-specific. A subject is excluded when a selected PD marker has no usable observation or a selected covariate has no usable value. An unusable PD observation alone does not exclude the subject if that same marker has another usable observation.

No observations for selected PD marker : pdresp

SUBJECTS 9 10

No observations for selected PD marker : pdcont

SUBJECTS 11

Run details

Running

run_01M0EM6MKH6RXGWBARBY30CTZW

PROGRESS

CREATED9 min ago

STARTED1 min ago

LAST UPDATEjust now

STAGES

✓ startup

→

✓ pk structure

→

pk final

→

pd final

→

publishing

INPUT

TIME BASIS
observed

PD MARKERS
pdcont Level expected to decrease with drug exposure
pdresp Level expected to increase with drug exposure

ALLOWED COVARIATES
weight age sex

Cancel run

EVENT LOG

All events Audit (user only)

model_build_run_creationAug 20, 06:42 AM GMT+3

succeededuser · Juho Timonenmodel_build_run: run_01M0EM6MKH6RXGWBARBY30CTZW

PROGRESS LOGS

pk final	Aug 20, 6:51:22 AM GMT+3	info	Fitting final PK model OneCompAbsorptionFirstOrder+Covariates - base configuration.
pk final	Aug 20, 6:51:21 AM GMT+3	info	Computing pooled initialization for final PK model OneCompAbsorptionFirstOrder+Covariates.
pk final	Aug 20, 6:51:21 AM GMT+3	info	Fitting final PK model OneCompAbsorptionFirstOrder+Covariates.
pk final	Aug 20, 6:51:20 AM GMT+3	info	Preparing final PK model covariates.
pk structure	Aug 20, 6:51:11 AM GMT+3	info	Comparing PK structure candidates.
pk structure	Aug 20, 6:51:11 AM GMT+3	info	Accepted base configuration for PK structure candidate TwoCompAbsorptionFirstOrder (max R-

Model fits

Selected PK structure: OneCompAbsorptionFirstOrder

Rationale: OneCompAbsorptionFirstOrder ranked best by conditional-psis-loo; TwoCompAbsorptionFirstOrder was not selected by this comparison.

PK structure candidates

FIT	STATUS		MODEL		INFERENCE	
PK structure candidate 1 mf_b149d638fc0048148e575001d5b0988d	FIT	Succeeded	STRUCTURE	OneCompAbsorptionFirstOrder	Final configuration base	Max R-hat 1.031
	POSTPROC	Succeeded	OBSERVATION	CensoredLogNormal		
PK structure candidate 2 mf_0a47d5dc9aa94a6c8c10dc4840e25e0c	FIT	Succeeded	STRUCTURE	TwoCompAbsorptionFirstOrder	Final configuration base	Max R-hat 1.037
	POSTPROC	Succeeded	OBSERVATION	CensoredLogNormal		



Final PK

FIT	STATUS		MODEL		INFERENCE	
PK mf_9df288e785f24d228e72c384f79710f1	FIT	Succeeded	STRUCTURE	OneCompAbsorptionFirstOrder	Final configuration fixed_mu=all_pk; recentered_fixed_mu=cl,v,ka_raw	Max R-hat 1.020
	POSTPROC	Succeeded	OBSERVATION	CensoredLogNormal		
			COVARIATES	age, sex, weight		



Final PD

FIT	STATUS		MODEL		INFERENCE	
PD: pdcont mf_f1409e2eb6ba4a709386595993645be7	FIT	Succeeded	STRUCTURE	EmaxEffectCompartment	Final configuration base	Max R-hat 1.070
	POSTPROC	Succeeded	OBSERVATION	CensoredNormal		
PD: pdresp mf_673c6c3e1bdb4532801f5e833715fee5	FIT	Succeeded	STRUCTURE	EmaxEffectCompartment	Final configuration fixed_mu=all_pd_except_e0; recentered_fixed_mu=ke0	Max R-hat 1.086
	POSTPROC	Succeeded	OBSERVATION	CensoredNormal		

Automatic PK fitting fallbacks

The ladder advances only while the current fit has not met the convergence target.

- 01 Initialize from a simpler fit shared across subjects
- 02 Fit the full PK model with between-subject variability
- 03 If close to convergence, run longer
- 04 If still not converged, fix population averages
- 05 If still not converged, remove subject variability one parameter at a time
- 06 If fixed averages are poorly centered, recenter and refit



Automatic PD fitting fallbacks

The ladder advances only while the current fit has not met the convergence target.

- 01 Use the selected PK fit as the exposure input
- 02 Fit the full effect-compartment Emax PD model with subject variability
- 03 If close to convergence, run longer
- 04 If still not converged, remove subject variability one term at a time
- 05 If still not converged, fix selected population averages and repeat
- 06 If fixed averages are poorly centered, recenter and refit



VeloMetrix and existing PK/PD workflows

Traditional projects

Connections between data, models, and reports are maintained project by project.

Established platforms

Flexible PK/PD modeling, diagnostics, and simulation.

NONMEM · MonolixSuite · Pumas · Stan/Torsten · nlmixr2

VeloMetrix

One application connects data preparation, analysis, model building, simulation, and reporting.



VeloMetrix aims to automate the workflow

